TabletCraft: Bridging a 4,000-Year Cultural Gap with Bidirectional Akkadian NMT and Cuneiform Rendering

Anonymous ACL submission

Abstract

Half a million cuneiform clay tablets survive in museums worldwide — yet modern humans can neither read nor write in the world's oldest writing system, creating a 4,000-year cultural barrier that existing NLP tools have only partially addressed. Prior work enables one-way, scholar-oriented translation from Akkadian to English, but offers no path in the reverse direction: ordinary people cannot express their own thoughts in cuneiform, and thus remain passive consumers of ancient culture rather than active participants. We present TABLETCRAFT, the first open-source system that enables *bidirectional* interaction with Mesopotamian writing. Users can read ancient tablets (Akkadian→English) and write their own messages as cuneiform clay tablets (English→Akkadian→cuneiform→rendered tablet). The system integrates a ByT5-based translation model trained on 116K bidirectional samples, a cuneiform sign converter with 14,240 mappings (95.3% coverage), and a visual tablet renderer — packaged as a `pip`-installable toolkit with CLI and web demo.

1 Introduction

Cuneiform is humanity's earliest writing system, spanning over three millennia (c. 3400 BCE – 75 CE) across ancient Mesopotamia (Walker, 1987). An estimated 500,000 clay tablets survive, recording royal decrees, epic poetry, trade receipts, and personal letters (Gutherz et al., 2023) — a cultural archive of extraordinary breadth. Yet this archive remains locked: fewer than a few hundred scholars worldwide can read cuneiform, and the majority of tablets remain untranslated (Gordin et al., 2020).

The resulting **cross-cultural barrier** is unlike any in contemporary NLP. It is not a barrier between two living communities who might learn each other's languages, but a *one-way temporal wall*: the voices of the world's first literate civilization can speak to us, but we cannot speak back. Existing NLP tools reinforce this asymmetry. Recent NMT systems (Gutherz et al., 2023; Chen et al., 2024) translate Akkadian *into* English, but no system translates English *into* Akkadian or renders the result in cuneiform. The cultural engagement they enable is therefore passive: scholars read ancient texts, but no one — scholar or otherwise — can write new ones.

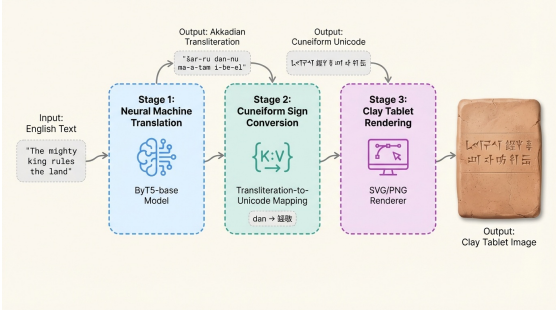

Figure 1: TABLETCRAFT pipeline: English text is translated to Akkadian, converted to cuneiform signs, and rendered as a clay tablet — enabling bidirectional cultural interaction.

To our knowledge, TABLETCRAFT is the first system that enables **bidirectional interaction** with ancient Mesopotamian writing. A student can type "The king rules the land" and receive a clay tablet bearing the cuneiform equivalent; a museum visitor can see their name in the world's oldest script; a researcher can back-translate English glosses into Akkadian for data augmentation. By making cuneiform not only readable but *writable*, we transform the relationship between modern users and ancient culture from consumption to exchange.

2 Cross-Cultural Design Challenges

Ancient-language NLP raises cross-cultural problems that mainstream multilingual NLP has not yet engaged with (Bird, 2020). We identify four challenges that shaped TABLETCRAFT's design.

Temporal distance as cultural distance. Akkadian translation bridges a civilizational gap, not merely a linguistic one. Concepts like "joint-stock capital," "eponym dating," or "divine determinatives" have no modern equivalents (Veenhof and Eidem, 2008). Our system preserves these culturally specific terms following Assyriological convention rather than forcing modernizing glosses.

The reverse direction as cultural agency. Prior systems treat ancient languages as objects of study. The English→Akkadian direction we introduce reframes users as *participants*: they can compose in cuneiform, fostering a personal connection that passive reading cannot (Luo et al., 2024). This comes with a risk of trivialization (Bender et al., 2021); we mitigate it by explicitly labeling all machine-generated Akkadian as approximate.

Accessibility vs. scholarly fidelity. We serve two populations — scholars needing precise transliterations and the public seeking approachable engagement (Terras, 2022) — by providing both raw transliteration and visual tablet outputs.

Whose heritage? Mesopotamian heritage is disproportionately studied in Western institutions (Rayne et al., 2017). An open-source, multilingual-extensible toolkit lowers barriers for researchers and educators in Iraq, Syria, Turkey, and Iran — the nations on ancient Mesopotamian land.

3 System Architecture

3.1 Translation Model

We fine-tune ByT5-base (Xue et al., 2022) (581M parameters) on 58,126 parallel sentences from Akkademia (Gutherz et al., 2023) (50K), a shared translation task (Deep Past Initiative, 2026) (1.5K), and sentence-aligned expansions (6K). We train bidirectionally (Ak→En and En→Ak with task prefixes), yielding 116K pairs (Sennrich et al., 2016). Byte-level tokenization handles diacritics (š, ṭ, ṣ) and logograms without vocabulary mismatch (Sahala et al., 2025). Training: 20 epochs, lr 10^{-4}, batch 32, label smoothing 0.2, BF16 on 2× RTX 3090.

3.2 Cuneiform Converter

A lookup table of 14,240 transliteration→Unicode cuneiform mappings compiled from ORACC (ORACC Project, 2024), CDLI (CDLI, 2024), and Akkademia. Determinatives are stripped per Assyriological convention (Huehnergard, 2011).

System	BLEU	chrF++
Akkademia CNN†	37.5	–
TABLETCRAFT (ByT5-base)	**49.1**	**63.1**

Table 1: Ak→En on the Akkademia validation set (2,812 Neo-Assyrian samples). †: Gutherz et al. (2023), evaluated on their test split (different domain mix).

3.3 Tablet Renderer

SVG/PNG output styled after Neo-Assyrian tablets (clay background, ruling lines, Noto Sans Cuneiform font). No GPU; <10ms per tablet.

4 Evaluation

Table 1 reports translation quality. Scores are on the Akkademia validation set, which consists primarily of Neo-Assyrian royal inscriptions; direct comparison with prior work on different test splits or text genres (e.g., Old Assyrian trade correspondence) requires caution (Veenhof and Eidem, 2008).

Domain gap as a cross-cultural lesson. Recent shared tasks on Akkadian NLP confirm that *text genre* is the dominant factor in translation quality: systems trained on royal inscriptions degrade substantially on commercial letters, even when supported by large bilingual dictionaries (Gordin et al., 2025). This mirrors a broader insight: cross-cultural NLP cannot treat "Akkadian" as a monolith any more than it can treat "English" as one — the cultural context of the source text matters as much as its language.

Cuneiform coverage. On 1,000 sampled transliterations from the Akkademia test set, the converter achieves 95.3% token coverage (93.8% logograms, 95.7% syllabic values).

Usage. TABLETCRAFT is `pip`-installable with CLI, Python API, and Gradio web demo. Use cases include **education** (students see their names in cuneiform (Wagensonner, 2019)), **museums** (interactive "write in cuneiform" exhibits), and **research** (back-translation for data augmentation (Gordin et al., 2025)).

5 Related Work

Gutherz et al. (2023) introduced Akkademia for Akkadian→English NMT. Gordin et al. (2020) applied word embeddings to cuneiform for cultural analysis. Recent work has applied ByT5 to cuneiform lemmatization (Sahala et al., 2025) and

LLMs to Akkadian NLP (Riemenschneider, 2025; Gordin et al., 2025). Yavasan and Gordin (2025) demonstrated T5 for Hittite cuneiform. ParsiPy (Farsi et al., 2025) provides NLP tools for historical Persian. In digital humanities, Terras (2022) and Luo et al. (2024) have argued for accessible tools bridging computational methods and cultural heritage. Our work differs by enabling *bidirectional* interaction — not only reading ancient texts but writing new ones in cuneiform — emphasizing cultural participation over passive analysis.

6 Conclusion

We presented TABLETCRAFT, a toolkit that transforms the relationship between modern users and ancient Mesopotamian culture from one-way reading to two-way interaction. By making cuneiform writable — not just readable — we open a channel for cultural engagement that 4,000 years of temporal distance had closed. Future work includes Sumerian support, cuneiform OCR, and multilingual extension to enable cross-cultural engagement from non-English-speaking communities.

Limitations

The En→Ak direction produces *approximate modern* transliterations, not authentic ancient text. Performance varies substantially across Akkadian dialects; in particular, Old Assyrian commercial texts differ in vocabulary, register, and formulaic patterns from the Neo-Assyrian training data. Even with 14,240 sign mappings and a 17K-lemma dictionary, domain-specific adaptation remains essential — large lexical resources cannot substitute for in-domain parallel data. The system currently supports only English.

Ethics Statement

This work aims to democratize access to ancient cultural heritage while respecting its scholarly context. Machine translations are labeled as approximations. We acknowledge that Mesopotamian heritage has special significance for the people of modern Iraq and neighboring nations, and advocate for their inclusion in developing such tools.

References

Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of FAccT*, pages 610–623.

Steven Bird. 2020. Decolonising speech and language technology. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 3504–3519.

CDLI. 2024. CDLI: Cuneiform digital library initiative.

Yanru Chen, Yongli Li, Benoît Vetter, and Adam Anderson. 2024. Machine learning for ancient languages: A survey. In *Proceedings of the 1st Workshop on Machine Learning for Ancient Languages (ML4AL 2024)*. Association for Computational Linguistics.

Deep Past Initiative. 2026. Deep past challenge: Translate Akkadian to English. Kaggle Competition.

Hadi Farsi and 1 others. 2025. ParsiPy: NLP toolkit for historical Persian texts in Python. In *Proceedings of the Second Workshop on Ancient Language Processing*. Association for Computational Linguistics.

Shai Gordin, Adam Anderson, and 1 others. 2025. Eva-Cun 2025 shared task: Lemmatization and token prediction in Akkadian and Sumerian using LLMs. In *Proceedings of the Second Workshop on Ancient Language Processing*. Association for Computational Linguistics.

Shai Gordin, Gai Gutherz, Ariel Elazary, Avital Romach, Enrique Jiménez, Jonathan Berant, and Luis Sáenz. 2020. Reading Akkadian cuneiform using natural language processing. *PLoS ONE*, 15(10):e0240511.

Gai Gutherz, Shai Gordin, Luis Sáenz, Omer Levy, and Jonathan Berant. 2023. Translating Akkadian to English with neural machine translation. *PNAS Nexus*, 2(5):pgad096.

John Huehnergard. 2011. *A Grammar of Akkadian*, 3rd edition. Eisenbrauns.

Jiwei Luo and 1 others. 2024. Digitization and NLP for cultural heritage: Opportunities and challenges. In *Proceedings of the Workshop on NLP for Digital Humanities*.

ORACC Project. 2024. ORACC: Open richly annotated cuneiform corpus.

Louise Rayne, Jennie Bradbury, David Mattingly, Graham Philip, Robert Bewley, and Andrew Wilson. 2017. Cultural heritage at risk in the 21st century. In *Endangered Archaeology in the Middle East and North Africa*. Oxford University Press.

Frederick Riemenschneider. 2025. Beyond base predictors: Using LLMs to resolve ambiguities in Akkadian lemmatization. In *Proceedings of the Second Workshop on Ancient Language Processing*. Association for Computational Linguistics.

Aleksi Sahala and 1 others. 2025. Lemmatization of cuneiform languages using the ByT5 model. In *Proceedings of the Second Workshop on Ancient Language Processing*. Association for Computational Linguistics.

Rico Sennrich, Barry Haddow, and Alexandra Birch. 2016. Improving neural machine translation models with monolingual data. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 86–96.

Melissa Terras. 2022. Digitisation and digital resources in the humanities. *Digital Scholarship in the Humanities*, 37(4):1153–1170.

Klaas R. Veenhof and Jesper Eidem. 2008. *Mesopotamia: The Old Assyrian Period*. Academic Press Fribourg.

Klaus Wagensonner. 2019. Digital cuneiform: Challenges and prospects. In *The Oxford Handbook of Cuneiform Culture*. Oxford University Press.

C. B. F. Walker. 1987. *Cuneiform*. University of California Press.

Linting Xue, Aditya Barua, Noah Constant, Rami Al-Rfou, Sharan Narang, Mihir Kale, Adam Roberts, and Colin Raffel. 2022. ByT5: Towards a token-free future with pre-trained byte-to-byte models. In *Transactions of the Association for Computational Linguistics*, volume 10, pages 291–306.

Murat Yavasan and Shai Gordin. 2025. From clay to code: Transforming Hittite texts for machine learning. In *Proceedings of the Second Workshop on Ancient Language Processing*. Association for Computational Linguistics.